

Strict Global PF₄ for the Riemann Kernel and a Continuous Fourier Separator

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Abstract

Let Φ be the classical Riemann, or de Bruijn–Newman, kernel. We prove that every translation minor of Φ of order at most four is strictly positive at strictly ordered nodes, and an exact origin-confluent calculation gives a negative order-five minor. Thus the exact global Polya-frequency order is four. For the Riemann-kernel positive theorem, a sweep-free two-mode argument proves global positivity of the log-curvature quantities q, F_2 and of the fully confluent invariant $\mathcal{C}_4$. Its finite certificate uses only rational polynomial algebra, Taylor inequalities, and geometric tails. The order-five origin certificate likewise uses exact rational arithmetic. A quotient-Wronskian argument reduces arbitrary fourth-order minors to $\partial_\xi \Psi < 0$. In the increasing curvature coordinate, its numerator is

$$\delta\Lambda \int_p^w W(t) \frac{\mathcal{C}_4(t)}{Q(t)^6 \kappa(t)^2} dt.$$

Here W is the difference of two explicitly normalized triangular CDFs. Its strict positivity follows from a one-crossing density ratio. The structural transport reductions and analytic sign estimates are displayed in the text and appendices; exact coefficient reconstructions remain explicitly identified certificate boundaries. We then construct an explicit positive even Schwartz kernel which is itself strictly PF₄ but whose entire Fourier transform has nonreal zeros; the latter sign follows from a short exact discriminant inequality. Thus continuous PF₃ and PF₄ membership alone do not force Fourier real-rootedness. The results are finite-order and have no implication for the Riemann Hypothesis.

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1 Introduction

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is PF_r when every translation-kernel minor

$$\det[f(x_i - y_j)]_{i,j=1}^k$$

is nonnegative for $1 \leq k \leq r$ and increasing node tuples. We call the property strict when the determinant is positive whenever both tuples are strictly increasing. This paper determines the exact finite Polya-frequency order of the Riemann kernel; see [1] for modern background and Schoenberg's characterization. Related finite-order and Riemann-kernel work is placed in Section 2.

Theorem 1.1 (Strict global PF_4). *For every $1 \leq k \leq 4$ and strictly increasing real tuples $x_1 < \dots < x_k$ and $y_1 < \dots < y_k$,*

$$\det[\Phi(x_i - y_j)]_{i,j=1}^k > 0.$$

A parity factorization of the fully confluent derivative matrix at the origin gives a negative order-five limit. Divided differences then transfer that sign to equally spaced translation minors at sufficiently small positive spacing; see Theorem 11.2. Hence Theorem 1.1 has the following exact consequence.

Corollary 1.2. *The exact global Polya-frequency order of Φ is four.*

The exact-order classification and the Fourier question are distinct. At the proved order-four level, the following example shows that finite-order membership alone does not force Fourier real-rootedness, even within the positive even Schwartz class.

Theorem 1.3 (Continuous PF_4 Fourier separator). *There is an explicit positive even Schwartz function f_* which is strict PF_4 , while its entire Fourier transform has nonreal zeros. Consequently continuous PF_3 and continuous PF_4 membership are each insufficient to force Fourier real-rootedness.*

The proof has two certified one-variable inputs and one exact transport mechanism. First, global inequalities $q > 0$ and $F_2 > 0$ imply strict PF_3 and positivity of a three-point functional Λ . Second, the fully confluent order-four invariant $\mathcal{C}_4$ is globally positive. Exact quotient identities reduce strict PF_4 to the monotonicity of a scalar three-point function Ψ . An increasing curvature coordinate then turns the derivative of Ψ into a positive crossing-kernel integral of $\mathcal{C}_4$.

The distinction between a human derivation and a computer certificate is maintained throughout. Directed interval arithmetic proves the compact and tail inequalities stated in Section 4; the structural transport algebra is written in the paper and independently replayed by repository verifiers. The exact separator coefficient list is printed in Appendix E. The short order-five certificate uses only exact rational arithmetic and elementary series bounds. The classification concerns finite-order total positivity only. It neither proves nor disproves the Riemann Hypothesis. Theorem 1.3, proved in Section 12, supplies an explicit order-four counterexample rather than a finite-order sufficiency threshold.

2 Relation to earlier work

We use *finite-order* Polya-frequency terminology in the sense reviewed by Belton, Guillot, Khare, and Putinar [1]. Khare’s characterization of measurable TN_p functions for $p \geq 3$ gives a broader finite-order context for translation-kernel criteria [2]; the present proof instead derives a kernel-specific strict order-four criterion through quotient Wronskians and a curvature transport identity.

For the same normalization of the Riemann kernel, Dimitrov and Xu relate Wronskians of Fourier and Laplace transforms to correlation kernels and real-zero criteria [3]. Csordas and Varga obtain strict concavity and moment inequalities for the Riemann kernel in connection with the Riemann Hypothesis [4]. Those works concern adjacent Wronskian, moment, and Fourier questions. Our theorem here is the finite translation-minor statement in Theorem 1.1; no comprehensive priority claim beyond that precise proved statement is made.

3 The kernel and smooth even continuation

Put

$$\vartheta(x) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 x}, \quad \Theta(t) = \vartheta(e^{2t}), \quad H(t) = e^{t/2} \Theta(t).$$

The theta transformation $\vartheta(x) = x^{-1/2}\vartheta(1/x)$ gives

$$\Theta(-t) = e^t \Theta(t), \quad H(-t) = H(t).$$

The series and all its derivatives converge locally uniformly, so H is real analytic and even. Define

$$\Phi(t) = \frac{1}{2} \left(H''(t) - \frac{1}{4} H(t) \right).$$

Termwise differentiation gives, for every real t ,

$$\Phi(t) = \sum_{n \geq 1} \left(4\pi^2 n^4 e^{9t/2} - 6\pi n^2 e^{5t/2} \right) e^{-\pi n^2 e^{2t}}. \quad (3.1)$$

Thus the positive-side formula often written with $|t|$ is the restriction of a real-analytic even function. In particular, smoothness at the origin is not inferred from numerical coverage.

For $t \geq 0$, the coefficient of the n -th exponential can be factored as

$$2\pi n^2 e^{5t/2} (2\pi n^2 e^{2t} - 3) > 0,$$

because $2\pi > 3$. Hence every term in (3.1) is positive and $\Phi(t) > 0$ on $[0, \infty)$. Evenness gives $\Phi(t) > 0$ on all of $\mathbb{R}$, so the following logarithm is globally defined.

Write

$$\ell = \log \Phi, \quad s = \ell', \quad q = -s' = -\ell''.$$

For $a < b$, define

$$A(a, b) = s(a) - s(b), \quad M(a, b) = \frac{q(b) - q(a)}{A(a, b)}. \quad (3.2)$$

When $q > 0$, s is strictly decreasing and $A(a, b) > 0$.

4 Certified one-variable inputs

Define

$$F_1 = qq'' - (q')^2, \quad F_2 = q^3 - F_1.$$

For the logarithmic cumulants $c_j = \ell^{(j)}$, set

$$\begin{aligned} m_0 &= 1, & m_1 &= 0, & m_2 &= c_2, & m_3 &= c_3, \\ m_4 &= c_4 + 3c_2^2, & m_5 &= c_5 + 10c_2c_3, \\ m_6 &= c_6 + 15c_2c_4 + 10c_3^2 + 15c_2^3, \end{aligned}$$

and

$$\mathcal{C}_4(t) = \det[m_{i+j-2}(t)]_{i,j=1}^4. \quad (4.1)$$

Lemma 4.1 (Certified input). *For the kernel Φ ,*

$$q(t) > 0, \quad F_2(t) > 0, \quad \mathcal{C}_4(t) > 0 \quad (t \in \mathbb{R}).$$

The proof is analytic and sweep-free. For $u \geq 0$, put

$$x = \pi e^{2u}, \quad y = e^{-3x}, \quad P_0(z) = 2z - 3,$$

and recursively

$$P_{j+1}(z) = \left(\frac{5}{2} - 2z\right) P_j(z) + 2zP_j'(z).$$

After division by the positive first theta mode, the first two normalized derivatives are

$$a_j(x, y) = \frac{P_j(x) + 4yP_j(4x)}{P_0(x)}.$$

Exact rational Bernstein coefficients on three fixed algebraic domains prove

$$q_{1,2} > 10, \quad (F_2)_{1,2} > 1000, \quad (\mathcal{C}_4)_{1,2} > 50,000,000.$$

The complete contribution of modes $n \geq 3$ is then bounded by one monotone polynomial-geometric estimate. Its changes in the cleared sign numerators are smaller than the three displayed margins, both on $\pi \leq x \leq 5$ and on $x \geq 5$. Evenness completes the real line. All inequalities, including the derivative paddings formerly used on the tail, are derived in Appendix B. The exact replay uses no floating point, no interval subdivision, and no FLINT or Arb arithmetic.

5 Order three and the positive slope functional

For $\xi < m < r$, define

$$\Lambda(\xi; m, r) = A(\xi, r) + M(\xi, m) - M(m, r). \quad (5.1)$$

We give the full estimate used to prove its positivity.

Set $f_0 = q'/q$. Because $s' = -q$,

$$M(a, b) = \frac{\int_a^b q'(t) dt}{\int_a^b q(t) dt} = \frac{\int_a^b f_0(t) q(t) dt}{\int_a^b q(t) dt};$$

thus $M(a, b)$ is a q -weighted mean of f_0 . Choose a point $u \in [\xi, m]$ where f_0 is minimal on the left interval and a point $v \in [m, r]$ where it is maximal on the right. Then

$$M(m, r) - M(\xi, m) \leq f_0(v) - f_0(u) \leq \int_{\xi}^r \max(f_0'(t), 0) dt.$$

Since

$$f_0' = \frac{qq'' - (q')^2}{q^2} = \frac{F_1}{q^2}$$

and $A(\xi, r) = \int_{\xi}^r q(t) dt$, equation (5.1) yields

$$\Lambda(\xi; m, r) \geq \int_{\xi}^r \left(q - \frac{\max(F_1, 0)}{q^2} \right) dt \quad (5.2)$$

$$= \int_{\xi}^r \frac{q^3 - \max(F_1, 0)}{q^2} dt = \int_{\xi}^r \frac{\min(q^3, F_2)}{q^2} dt > 0. \quad (5.3)$$

The last inequality follows from Lemma 4.1 and $q > 0$.

The same quantity is the logarithmic slope derivative in the normalized order-three determinant. Hence (5.3), together with strict log-concavity, proves strict global PF₃. We rederive the needed Wronskian form in the next section because its signs also control order four.

6 Quotient-Wronskian reduction

Fix $y_1 < y_2 < y_3 < y_4$, put $u_j(t) = \Phi(t - y_j)$, and write $p_j = t - y_j$, so $p_4 < p_3 < p_2 < p_1$. Define

$$v_j = (u_j/u_1)', \quad w_j = (v_j/v_2)'.$$

Successive column division and differentiation give

$$W(u_1, u_2, u_3) = u_1^3 W(v_2, v_3), \quad (6.1)$$

$$W(u_1, u_2, u_3, u_4) = u_1^4 v_2^3 w_3^2 \frac{d}{dt} \left(\frac{w_4}{w_3} \right). \quad (6.2)$$

Let $\mathcal{T}$ denote simultaneous translation of all point arguments. From (3.2),

$$\mathcal{T}A(a, b) = q(b) - q(a), \quad \mathcal{T} \log A(a, b) = M(a, b). \quad (6.3)$$

This sign is load-bearing. Since

$$\frac{v_j}{v_2} = \frac{u_j}{u_2} \frac{A(p_j, p_1)}{A(p_2, p_1)},$$

direct differentiation gives

$$g_j := \mathcal{T} \log(v_j/v_2) = A(p_j, p_2) + M(p_j, p_1) - M(p_2, p_1). \quad (6.4)$$

The exact weighted-mean split

$$A(p_j, p_1)M(p_j, p_1) = A(p_j, p_2)M(p_j, p_2) + A(p_2, p_1)M(p_2, p_1)$$

turns (6.4) into

$$g_j = \frac{A(p_j, p_2)}{A(p_j, p_1)} \Lambda(p_j; p_2, p_1) > 0. \quad (6.5)$$

Thus $w_j = (v_j/v_2)g_j > 0$, and (6.1) has the strict order-three sign.

Define

$$\Psi(\xi; m, r) = \Lambda(\xi; m, r) + \mathcal{T} \log \Lambda(\xi; m, r).$$

Because $w_j = (v_j/v_2)g_j$,

$$\mathcal{T} \log(w_4/w_3) = (g_4 + \mathcal{T} \log g_4) - (g_3 + \mathcal{T} \log g_3). \quad (6.6)$$

Equation (6.5) and (6.3) give

$$\mathcal{T} \log g_j = M(p_j, p_2) - M(p_j, p_1) + \mathcal{T} \log \Lambda_j,$$

where $\Lambda_j = \Lambda(p_j; p_2, p_1)$. A second use of the same mean split gives

$$g_j + M(p_j, p_2) - M(p_j, p_1) = \Lambda_j - A(p_2, p_1).$$

The common last term cancels in (6.6), yielding the exact identity

$$\mathcal{T} \log(w_4/w_3) = \Psi(p_4; p_2, p_1) - \Psi(p_3; p_2, p_1). \quad (6.7)$$

Proposition 6.1 (Three-point equivalence). *Under the strict lower-order positivity already proved, global PF₄ is equivalent to*

$$\partial_\xi \Psi(\xi; m, r) \leq 0 \quad (\xi < m < r).$$

Strict inequality implies strict global PF₄.

Proof. If $\partial_\xi \Psi \leq 0$, then $p_4 < p_3$ makes the right side of (6.7) nonnegative. All prefactors in (6.2) are positive. The Wronskian is therefore nonnegative, and is positive in the strict case.

For increasing row nodes $t_1 < \dots < t_k$, divide columns by $u_1(t_i)$ and take consecutive row differences from the bottom upward. The fundamental theorem of calculus and determinant multilinearity give

$$\det[U_j(t_i)]_{i,j=1}^k = \int_{t_1}^{t_2} \dots \int_{t_{k-1}}^{t_k} \det[U'_{j+1}(\tau_i)]_{i,j=1}^{k-1} d\tau_1 \dots d\tau_{k-1},$$

where $U_1 = 1$. Repeating the quotient reduction converts the Wronskian sign to every collocation minor; positivity of the integrand on a product of positive-length intervals gives the strict conclusion.

Conversely, assume global PF₄. Let all four row nodes coalesce, divide by their positive Vandermonde, and pass to the limit. The resulting translate Wronskian is nonnegative. Equations (6.2) and (6.7) imply $\Psi(p_4; p_2, p_1) \geq \Psi(p_3; p_2, p_1)$ whenever $p_4 < p_3$. Thus $\Psi(\cdot; m, r)$ is nonincreasing, which is equivalent to the stated differential inequality because it is smooth. $\square$

7 Curvature coordinate and sign bridge

Since $q > 0$,

$$y(t) = -s(t), \quad Q(y(t)) = q(t)$$

defines a global strictly increasing coordinate on its image, with $dy/dt = q(t) = Q(y) > 0$. Primes in this and the next three sections mean y -derivatives. Put

$$\rho = 1 - Q'' = \frac{F_2}{q^3} > 0, \quad \kappa = 2 - Q'' = 1 + \rho > 1.$$

For $\xi < m < r$, write

$$p = y(\xi), \quad z = y(m), \quad w = y(r), \quad L = z - p, \quad R = w - z.$$

Because $A(a, b) = y(b) - y(a)$ and $M(a, b) = Q[y(a), y(b)]$,

$$\Lambda = (w - p) + Q[p, z] - Q[z, w]. \quad (7.1)$$

Two integrations of $\kappa = 2 - Q''$ give

$$\Lambda = \frac{1}{L} \int_p^z (y - p) \kappa(y) dy + \frac{1}{R} \int_z^w (w - y) \kappa(y) dy, \quad (7.2)$$

$$\delta := -\partial_p \Lambda = \frac{1}{L^2} \int_p^z (z - y) \kappa(y) dy > 0. \quad (7.3)$$

Simultaneous translation becomes

$$\mathcal{T} = Q(p) \partial_p + Q(z) \partial_z + Q(w) \partial_w,$$

and $\partial_\xi = Q(p) \partial_p$. Thus $\Lambda_\xi = -Q(p) \delta$.

Define

$$\mathcal{N} = \delta \Lambda^2 + Q'(p) \delta \Lambda + \Lambda \mathcal{T} \delta - \delta \mathcal{T} \Lambda. \quad (7.4)$$

Differentiating $\Psi = \Lambda + \mathcal{T} \log \Lambda$, commuting simultaneous translation with ∂_ξ , and collecting over Λ^2 gives

$$\partial_\xi \Psi = -\frac{Q(p)}{\Lambda^2} \mathcal{N}. \quad (7.5)$$

Equivalently,

$$\frac{\mathcal{N}}{\delta\Lambda} = K := \Lambda + Q'(p) + \mathcal{T} \log \delta - \mathcal{T} \log \Lambda. \quad (7.6)$$

It remains to prove $K > 0$.

8 The confluent invariant in curvature form

Define

$$D = 3(\kappa - 1) - \{Q(\log \kappa)'\}' . \quad (8.1)$$

Lemma 8.1 (Confluent-to-curvature identity). *The determinant (4.1) satisfies*

$$\mathcal{C}_4 = Q^6 \kappa^2 D. \quad (8.2)$$

Consequently $D > 0$ on $\mathbb{R}$.

Proof. The change of variable gives $d/dt = Q d/dy$ and $c_2 = -Q$. Recursively applying this operator produces the five cumulants listed in Appendix A. Substitution into the Schur-complement form of (4.1) factors out Q^6 . The remaining factor is

$$\kappa^2 (3(\kappa - 1) - \{Q(\log \kappa)'\}'),$$

as shown by the common expanded numerator in Appendix A. This proves (8.2) without a numerical assumption. Lemma 4.1, $Q > 0$, and $\kappa > 0$ then imply $D > 0$. $\square$

The determinant definition is primary: it is the doubly confluent fourth-order translation minor after division by the positive row and column Vandermonde factors and by Φ^4 . The explicit thirteen-term expansion is included in Appendix A for independent checking of the certificate input.

9 Exact transport identity

Normalize the triangular weights in (7.2) and (7.3) as probability measures:

$$d\mu(y) = \frac{(z - y)\kappa(y)}{L^2\delta} \mathbf{1}_{[p,z]}(y) dy, \quad (9.1)$$

$$d\nu(y) = \frac{(y - p)\kappa(y)}{L\Lambda} \mathbf{1}_{[p,z]}(y) dy + \frac{(w - y)\kappa(y)}{R\Lambda} \mathbf{1}_{[z,w]}(y) dy. \quad (9.2)$$

Set

$$A_0(y) = 3(y - Q'(y)) - Q(y) \frac{\kappa'(y)}{\kappa(y)}.$$

Then $A'_0 = D$.

Lemma 9.1 (Transport cancellation). *With K as in (7.6),*

$$K = \mathbb{E}_\nu[A_0] - \mathbb{E}_\mu[A_0]. \quad (9.3)$$

Proof. Two elementary primitives carry the complete cancellation:

$$\kappa A_0 = H', \quad H = 3y^2 - 3Q - 3yQ' + (Q')^2 + QQ'', \quad (9.4)$$

$$H = J', \quad J = y^3 - 3yQ + QQ'. \quad (9.5)$$

Both identities follow by one differentiation.

Let $I_L = J(z) - J(p)$ and $I_R = J(w) - J(z)$. Integrating the linear weights in (9.1)–(9.2) by parts once gives

$$\mathbb{E}_\nu[A_0] - \mathbb{E}_\mu[A_0] = \frac{-I_L/L + I_R/R}{\Lambda} + \frac{LH(p) - I_L}{L^2\delta}. \quad (9.6)$$

On the other hand, differentiation of the triangular integrals along $\mathcal{T}$ gives

$$\mathcal{T}\Lambda = U(z, w) - U(p, z), \quad \mathcal{T}\delta = \frac{U(p, z) - (Q(z) - Q(p))\delta - Q(p)\kappa(p)}{L},$$

where

$$U(c, d) = Q(c) + Q(d) - \frac{Q(d)Q'(d) - Q(c)Q'(c)}{d - c} + \frac{(Q(d) - Q(c))^2}{(d - c)^2}. \quad (9.7)$$

Substitution in (7.6) yields

$$K = \Lambda + Q'(p) - Q[p, z] + \frac{U(p, z) - U(z, w)}{\Lambda} + \frac{U(p, z) - Q(p)\kappa(p)}{L\delta}. \quad (9.8)$$

The endpoint identity equating (9.6) and (9.8), with every substitution exposed, is Appendix C. It is an identity in the endpoint positions and the values of Q, Q', Q'' ; no bound or polynomial model is used. This proves (9.3). $\square$

10 The positive crossing kernel

Let F_μ, F_ν be the CDFs of (9.1)–(9.2), and put

$$W(y) = F_\mu(y) - F_\nu(y).$$

Lemma 10.1 (Strict stochastic ordering). *One has $W(y) > 0$ for $p < y < w$, while $W(p) = W(w) = 0$.*

Proof. On $[z, w]$, $F_\mu = 1$, so $W = 1 - F_\nu > 0$ before w . On (p, z) , the ratio of the two positive densities is

$$R_0(y) = \frac{d\mu/dy}{d\nu/dy} = \frac{\Lambda(z - y)}{L\delta(y - p)}. \quad (10.1)$$

Writing $C = \Lambda/(L\delta) > 0$, direct differentiation gives

$$R'_0(y) = -\frac{C(z - p)}{(y - p)^2} < 0. \quad (10.2)$$

Thus R_0 decreases strictly from $+\infty$ to 0. Hence $W' = \mu' - \nu'$ changes sign exactly once, from positive to negative. Since $W(p) = 0$ and

$$W(z) = 1 - F_\nu(z) = \nu((z, w]) > 0,$$

the function cannot return to zero on $(p, z]$. The right interval was already handled. $\square$

Integration by parts for CDFs, using $A'_0 = D$ and the common total mass, turns Lemma 9.1 into

$$K = \int_p^w W(y)D(y)dy. \quad (10.3)$$

Combining (7.6), (8.2), and (10.3) gives the central identity

$$\mathcal{N} = \delta\Lambda \int_p^w W(y) \frac{\mathcal{C}_4(y)}{Q(y)^6 \kappa(y)^2} dy > 0. \quad (10.4)$$

Every factor in the interior is strictly positive.

11 Completion and exact order

Equation (7.5) and (10.4) imply

$$\partial_\xi \Psi(\xi; m, r) < 0 \quad (\xi < m < r).$$

Proposition 6.1 therefore proves Theorem 1.1.

We now prove directly that the classification stops at order four. Put

$$H_k = \det[\Phi^{(i+j)}(0)]_{i,j=0}^{k-1}, \quad s_j = \Phi^{(j)}(0).$$

Since Φ is even, simultaneous parity permutations of the rows and columns give

$$H_4 = AB, \quad H_5 = CB,$$

where

$$A = s_0 s_4 - s_2^2, \quad B = s_2 s_6 - s_4^2,$$

and

$$C = \det \begin{pmatrix} s_0 & s_2 & s_4 \\ s_2 & s_4 & s_6 \\ s_4 & s_6 & s_8 \end{pmatrix}.$$

Lemma 11.1. *For the Riemann kernel,*

$$445 < A < 446, \quad 189223 < B < 189228, \quad -674000000 < C < -614000000.$$

In particular, $H_4 > 0$ and $H_5 < 0$.

Proof. For $x = \pi n^2 e^{2u}$, the n -th summand of $\Phi(u)$ is $e^{u/2-x} P_0(x)$, where $P_0(x) = 2x(2x-3)$. If its j -th derivative is $e^{u/2-x} P_j(x)$, then

$$P_{j+1} = 2xP_j' + (1/2 - 2x)P_j.$$

Thus

$$s_j = \sum_{n \geq 1} e^{-\pi n^2} P_j(\pi n^2).$$

The first three modes are enclosed by rational alternating-series bounds. Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ supplies rational bounds for π , and argument division followed by the degree-29 and degree-30 alternating Taylor sums supplies rational bounds for each exponential.

For the omitted modes, let S_j be the sum of the absolute coefficients of P_j . Since $\deg P_j = j+2$, $3 < \pi < 22/7$, and $e^3 > \sum_{r=0}^8 3^r/r! > 20$,

$$\sum_{n \geq 4} |P_j(\pi n^2)| e^{-\pi n^2} \leq \frac{S_j (22/7)^{j+2} 4^{2j+4} 20^{-16}}{1 - (5/4)^{20} 20^{-9}} \quad (j = 0, 2, 4, 6, 8).$$

All endpoints lie on $10^{-36}\mathbb{Z}$, and every arithmetic operation is rounded outward. Substitution into the displayed formulas gives the entirely rational sign chain

$$0 < 445 < A < 446, \quad 0 < 189223 < B < 189228,$$

$$-674000000 < C < -614000000 < 0.$$

Thus $AB > 0$ and $CB < 0$ without a floating-point determinant. The full grid endpoints and the standard-library verifier are included in the reproducibility archive. $\square$

Theorem 11.2. *The Riemann kernel is not PF_5 .*

Proof. For a C^{2k-2} function f , repeated row and column divided differences give, for increasing fixed tuples a, b ,

$$\lim_{h \downarrow 0} \frac{\det[f(z + h(a_i - b_j))]_{i,j=0}^{k-1}}{h^{k(k-1)} V(a) V(b)} = \frac{(-1)^{k(k-1)/2}}{\prod_{r=0}^{k-1} (r!)^2} \det[f^{(i+j)}(z)]_{i,j=0}^{k-1}.$$

Take $k = 5$, $z = 0$, and $a_i = b_i = i$. The sign factor is one, both Vandermonde factors are positive, and Lemma 11.1 makes the limit strictly negative. Hence $\det[\Phi((i-j)h)]_{i,j=0}^4 < 0$ for every sufficiently small positive h . These are translation minors at strictly increasing nodes. $\square$

Theorem 1.1 and Theorem 11.2 prove Corollary 1.2.

The proof uses no multidimensional interval cover of the nonconfluent configuration space. Earlier positive-tail Peano densities, collision cones, Hermite boxes, and escape charts remain independent certified results, but they are not premises here. Their role was superseded when the transport numerator was recognized as the difference of expectations in (9.3).

Remark 11.3 (Riemann Hypothesis boundary). The theorem determines the kernel's exact finite Polya-frequency order and makes no claim about the zeros of the Riemann Ξ -function. In particular, the failure of PF_5 neither proves nor disproves the Riemann Hypothesis. The next section proves, by an explicit strict- PF_4 separator, that no Fourier real-rootedness conclusion follows from membership through order four alone.

12 An explicit continuous PF_4 separator

We now prove Theorem 1.3. Put

$$c = \frac{1}{64}, \quad B(w) = \sum_{k=-3}^3 b_k w^k, \quad (b_{-3}, \dots, b_3) = (2, 10, 23, 30, 23, 10, 2),$$

and define

$$f_*(x) = e^{-cx^2} B(e^{2cx}) = \sum_{k=-3}^3 b_k e^{ck^2} e^{-c(x-k)^2}. \quad (12.1)$$

The Gaussian-mixture expression proves positivity and Schwartz decay; palindromy of (b_k) proves evenness.

12.1 Strict PF_3

For fixed x , give $K \in \{-3, \dots, 3\}$ probability proportional to $b_k e^{2ckx}$, and write κ_j for its cumulants. Differentiation of the finite log-partition function gives

$$q = 2c - (2c)^2 \kappa_2, \quad q' = -(2c)^3 \kappa_3, \quad q'' = -(2c)^4 \kappa_4. \quad (12.2)$$

If $\mu = \mathbb{E}K$, then $0 \leq \mathbb{E}[(K+3)(3-K)] = 9 - \kappa_2 - \mu^2$. Thus $0 \leq \kappa_2 \leq 9$, and hence

$$q \geq 2c(1 - 18c) = 2c \frac{23}{32} > 0. \quad (12.3)$$

Moreover,

$$\begin{aligned} F_2 &= q^3 - \{qq'' - (q')^2\} \\ &= q^3 + q(2c)^4\kappa_4 + (2c)^6\kappa_3^2. \end{aligned} \quad (12.4)$$

The identity $\kappa_4 = \mathbb{E}(K - \mathbb{E}K)^4 - 3\kappa_2^2$ gives $\kappa_4 \geq -243$. Dropping the final nonnegative term in (12.4) gives

$$F_2 \geq q\{q^2 - 3888c^4\}.$$

At $c = 1/64$, the two coefficients after division by c^2 are

$$q^2/c^2 \geq \frac{529}{256}, \quad 3888c^2 = \frac{243}{256}.$$

Thus $F_2 \geq qc^2(143/128) > 0$ globally. The weighted-mean criterion of Section 5 proves strict PF₃.

12.2 Exact order-four density

Let $\epsilon = 2c = 1/32$, $t = e^{\epsilon x}$, and

$$P(t) = t^3 B(t) = 2 + 10t + 23t^2 + 30t^3 + 23t^4 + 10t^5 + 2t^6.$$

For $E = t \frac{d}{dt}$, define

$$N_1 = tP', \quad N_{j+1} = t(N_j'P - jN_jP'). \quad (12.5)$$

The quotient rule gives $E^j \log P = N_j/P^j$. Therefore, for $\ell_j = (\log f_*)^{(j)}$,

$$\ell_2 = \frac{-\epsilon P^2 + \epsilon^2 N_2}{P^2}, \quad \ell_j = \frac{\epsilon^j N_j}{P^j} \quad (3 \leq j \leq 6). \quad (12.6)$$

Insert these jets into the central-moment determinant (4.1). Every monomial has derivative weight twelve, so the common denominator is P^{12} :

$$\mathcal{C}_4[f_*](x) = \frac{H(t)}{P(t)^{12}}. \quad (12.7)$$

The exact expansion printed in Appendix E proves that H is palindromic of degree 72 and that all 73 coefficients are strictly positive. Hence $\mathcal{C}_4[f_*] > 0$ on $\mathbb{R}$.

The established inequalities $q > 0$ and $F_2 > 0$ also give

$$\kappa = 2 - Q'' = \frac{q^3 + F_2}{q^3} > 0.$$

The generic transport identity (10.4) now has strictly positive density and crossing weight for f_* . Thus $\partial_\epsilon \Psi < 0$, and Proposition 6.1 proves that f_* is strict PF₄.

12.3 Nonreal Fourier zeros

With $\widehat{f}(z) = \int_{\mathbb{R}} f(x)e^{-izx} dx$, Gaussian translation gives

$$\widehat{f}_*(z) = \sqrt{\frac{\pi}{c}} e^{-z^2/(4c)} \sum_{k=-3}^3 b_k e^{ck^2} e^{-ikz}. \quad (12.8)$$

Put $w = e^{-iz}$ and $u = w + w^{-1}$. The Laurent factor in (12.8) vanishes exactly when

$$R_c(u) = 2e^{9c}u^3 + 10e^{4c}u^2 + (23e^c - 6e^{9c})u + 30 - 20e^{4c} \quad (12.9)$$

vanishes. At $c = 1/64$, the exact calculation in Appendix E gives $\text{disc}(R_c) < 0$. Thus the real cubic has a nonreal conjugate pair. For $|w| = 1$, however, $w + w^{-1}$ is real. The corresponding w -roots therefore lie off the unit circle and lift through $w = e^{-iz}$ to zeros with $\text{Im } z \neq 0$. The Gaussian factor is zero-free. This completes the proof of Theorem 1.3.

13 Reproducibility and evidence boundary

The maintained paper has five active replay boundaries.

Object	Repository verifier	Role
Global q, F_2, C_4	<code>scripts/verify_riemann_signs_exact.py</code>	exact analytic inequalities
Quotient reduction	<code>scripts/verify_pf4_wronskian_reduction.py</code>	exact generic algebra
Transport completion	<code>scripts/verify_pf4_transport_kernel.py</code>	exact algebra and checks
Origin order-five obstruction	<code>scripts/verify_pf5_origin.py</code>	exact rational enclosures
Continuous strict-PF ₄ separator	<code>scripts/verify_continuous_pf4_separator.py</code>	exact polynomial algebra

The global-sign verifier retains two theta modes exactly, checks 361 rational Bernstein or decay coefficients and 140 derivative-envelope coefficients, and bounds all modes $n \geq 3$ by the analytic estimates in Appendix B. Its launcher forces SymPy’s pure-Python rational domain and rejects every attempted FLINT import. There is no interval subdivision and no floating-point sign decision.

The origin order-five certificate is independent. It uses only standard-library integer arithmetic with outward rounding on a 10^{-36} -rational grid, Machin’s formula, alternating Taylor sums, three retained theta modes, and one geometric tail bound. The separator certificate reconstructs all 73 positive coefficients of its order-four numerator and checks the discriminant inequality in Appendix E over the exact rationals.

The quotient, curvature, primitive, endpoint, density-ratio, and confluent-limit statements are mathematical identities for generic smooth functions. The paper displays their derivations; the symbolic scripts replay the algebra. High-precision values at $t = 1/3, 1, 4$ and several Riemann-kernel points are regression checks only and are not premises of a sign conclusion.

The former 192-bit Arb covers of 7731 and 8050 cells are retained in the repository as historical certificates, but they are archived and are not dependencies of this paper or of routine replay. The collision, Hermite, escape, and spatial-atlas computations are likewise outside the proof graph.

A focused independent audit should proceed in this order:

1. verify the analytic theta normalization in (3.1);
2. check (B.1), the two-mode Bernstein domains, and the later-mode estimates in Appendix B;
3. check the two weighted-mean splits in Section 5;
4. expand the determinant and curvature identity in Appendix A;
5. check the endpoint identity in Appendix C;

6. differentiate the density ratio in (10.2);
7. derive the parity factorization and divided-difference limit in Section 11, then replay the rational origin certificate;
8. reconstruct the separator determinant and exact discriminant proof in Appendix E.

No numerical scan is used as evidence for global nonconfluent positivity of either kernel.

The non-mutating entry point `scripts/replay_paper.py` checks all five registered verifier outputs. Exact package versions are in `requirements-paper.txt`; `python-flint` is not a paper dependency.

14 Declarations, availability, and computational provenance

Author contribution. Joshua Rainstar conceived and directed the project, developed the proof architecture, assembled and audited the computational evidence, and prepared the manuscript.

Acknowledgments and computational assistance. The author acknowledges the use of OpenAI Codex and other disclosed AI systems for symbolic derivation, implementation, drafting, and adversarial audit. The proof records preserve the resulting calculations and their replay boundaries; the author takes responsibility for the mathematical claims.

Conflict of interest. The author declares no competing interests.

Code and data availability. The public repository is <https://github.com/falseywinchnet/zeta>. The sweep-free analytic advance audited in this revision is frozen at MIND epoch P000072, Git commit 66ba7f983a21. Its audited paper integration is MIND epoch P000073. A submission release must freeze this revised source, its generated coefficient table, replay tool, environment lock, and built PDF under one versioned archive record; the PDF hash belongs in that release manifest rather than recursively inside the PDF.

All theorem-bearing sign decisions are exact. Numerical values produced by the repository scripts are diagnostic only; there is no separate experimental dataset.

References

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A Confluent and curvature algebra

The central-moment Hankel matrix is

$$\begin{pmatrix} 1 & 0 & m_2 & m_3 \\ 0 & m_2 & m_3 & m_4 \\ m_2 & m_3 & m_4 & m_5 \\ m_3 & m_4 & m_5 & m_6 \end{pmatrix}.$$

Taking the Schur complement of the upper-left entry reduces its determinant to

$$\det \begin{pmatrix} m_2 & m_3 & m_4 \\ m_3 & m_4 - m_2^2 & m_5 - m_2 m_3 \\ m_4 & m_5 - m_2 m_3 & m_6 - m_3^2 \end{pmatrix}.$$

Substitution of the moments from Section 4 and ordinary three-by-three expansion gives

$$\mathcal{C}_4 = 12c_2^6 + 24c_2^4c_4 - 24c_2^3c_3^2 + 2c_2^3c_6 - 12c_2^2c_3c_5 + 7c_2^2c_4^2 \quad (\text{A.1})$$

$$+ 12c_2c_3^2c_4 + c_2c_4c_6 - c_2c_5^2 - 9c_3^4 - c_3^2c_6 + 2c_3c_4c_5 - c_4^3. \quad (\text{A.2})$$

In the curvature coordinate, repeated application of $Q \, d/dy$ to $c_2 = -Q$ gives

$$\begin{aligned} c_2 &= -Q, \\ c_3 &= -QQ', \\ c_4 &= -Q\{(Q')^2 + QQ''\}, \\ c_5 &= -Q\{(Q')^3 + 4QQ'Q'' + Q^2Q'''\}, \\ c_6 &= -Q\{(Q')^4 + 11Q(Q')^2Q'' + 4Q^2(Q'')^2 + 7Q^2Q'Q''' + Q^3Q''''\}. \end{aligned}$$

Substitution in the preceding polynomial gives

$$\mathcal{C}_4 = -Q^6\mathcal{P},$$

where

$$\begin{aligned} \mathcal{P} &= QQ''Q'''' - Q(Q''')^2 - 2QQ'''' + Q'Q''Q''' - 2Q'Q'' \\ &\quad + 3(Q'')^3 - 15(Q'')^2 + 24Q'' - 12. \end{aligned}$$

Now $\kappa = 2 - Q''$, and direct differentiation gives

$$3(\kappa - 1) - \{Q(\log \kappa)'\}' = -\frac{\mathcal{P}}{(Q'' - 2)^2} = -\frac{\mathcal{P}}{\kappa^2}.$$

Therefore $\mathcal{C}_4 = Q^6\kappa^2D$, proving (8.2) by an exposed common numerator.

B Sweep-free proof of the certified input

This appendix proves Lemma 4.1. The positive-side theta series has j -th derivative

$$2\pi n^2 e^{5u/2 - n^2 x} P_j(n^2 x), \quad x = \pi e^{2u},$$

where the polynomials P_j satisfy the recurrence in Section 4. Machin's identity and alternating series give $\pi > 157/50$; hence $P_0(x) = 2x - 3 > 0$.

B.1 Cleared sign polynomials

In addition to a_j , define the normalized remaining modes

$$e_j(x) = \sum_{n \geq 3} \frac{n^2 e^{-(n^2-1)x} P_j(n^2 x)}{P_0(x)}, \quad b_j = a_j + e_j.$$

For an indeterminate raw jet $(b_0, \dots, b_6)$, put

$$N_q = b_1^2 - b_0 b_2,$$

$$\begin{aligned} N_{F_2} = & -b_0^4 b_2 b_4 + b_0^4 b_3^2 + b_0^3 b_1^2 b_4 - 2b_0^3 b_1 b_2 b_3 + 2b_0^3 b_2^3 \\ & - 3b_0^2 b_1^2 b_2^2 + 3b_0 b_1^4 b_2 - b_1^6, \end{aligned}$$

and

$$N_{C_4} = \det[b_{i+j}]_{i,j=0}^3.$$

Direct logarithmic differentiation and the central-moment determinant give

$$q = \frac{N_q}{b_0^2}, \quad F_2 = \frac{N_{F_2}}{b_0^6}, \quad C_4 = \frac{N_{C_4}}{b_0^4}. \quad (\text{B.1})$$

The thirteen-term cumulant expansion obtained from the last identity agrees with (A.1); this is an exact polynomial identity.

For orientation, retaining only the first mode gives

$$\begin{aligned} q_1 &= \frac{4x(4x^2 - 12x + 15)}{(2x - 3)^2}, \\ (F_2)_1 &= \frac{64x^3(64x^6 - 576x^5 + 2448x^4 - 6432x^3 + 10044x^2 - 8100x + 2295)}{(2x - 3)^6}, \\ (C_4)_1 &= \frac{49152x^6(16x^4 - 96x^3 + 360x^2 - 840x + 945)}{(2x - 3)^4}. \end{aligned}$$

After $x = 5 + z$, every numerator has positive coefficients. The second mode is retained exactly because it supplies the larger margin near the origin.

B.2 Exact two-mode margin

Alternating Taylor bounds for the exponential imply

$$0 < y < \frac{1}{12000} \quad (x \geq \pi),$$

and

$$\frac{1}{23000} < y < \frac{1}{12000} \quad \left(\frac{157}{50} \leq x \leq \frac{10}{3} \right).$$

We use the following elementary certificate lemma. If $p(s, t) = \sum p_{ij} s^i t^j$ has bidegree (d, e) , its Bernstein coefficients on the unit square are

$$\beta_{k\ell} = \sum_{i \leq k, j \leq \ell} p_{ij} \frac{\binom{k}{i} \binom{\ell}{j}}{\binom{d}{i} \binom{e}{j}}.$$

The power-to-Bernstein identity proves that $\beta_{k\ell} > 0$ for every k, ℓ implies $p > 0$ on the square.

Apply this lemma after affine rational changes of variables. For q , one rectangle ends at $x = 10/3$, followed by a positive base polynomial minus exponentially decreasing corrections. For F_2 , dependent bounds on the first rectangle are followed by a rectangle through $x = 5$ and a positive half-strip expansion. For $\mathcal{C}_4$, the first rectangle is followed by two negative odd-power corrections whose ratios to the positive base decrease. The derivative of each ratio has a numerator with positive coefficients after the indicated shift. The 361 resulting rational coefficient inequalities give

$$q_{1,2} > 10, \quad (F_2)_{1,2} > 1000, \quad (\mathcal{C}_4)_{1,2} > 50,000,000. \quad (\text{B.2})$$

This is a finite algebraic proof on fixed domains, not an interval sweep.

B.3 All later modes

For $t \geq 27$, shifting $t = 27 + z$ gives positive coefficients in $(-1)^j P_j(t)$ and in $2^{j+1} t^{j+1} - (-1)^j P_j(t)$. Therefore

$$0 < (-1)^j P_j(t) \leq 2^{j+1} t^{j+1}, \quad 0 \leq j \leq 6. \quad (\text{B.3})$$

The derivative numerator of the normalized $n = 3$ term is another positive shifted polynomial, so its absolute value decreases for $x \geq 3$. For $n \geq 4$, successive majorants have ratio at most

$$\left(\frac{5}{4} \right)^{16} e^{-27} < 10^{-9}.$$

Rational Taylor lower bounds for e^{24}, e^{27}, e^{45} then yield, on $\pi \leq x \leq 5$,

$$(|e_0|, \dots, |e_6|) < (7 \cdot 10^{-9}, 4 \cdot 10^{-7}, 2 \cdot 10^{-4}, 7 \cdot 10^{-4}, 3 \cdot 10^{-2}, 1.1, 41). \quad (\text{B.4})$$

The same Bernstein lemma gives

$$(|a_0|, \dots, |a_6|) < (2, 5, 20, 200, 2000, 60000, 1000000).$$

Expanding the three polynomials in (B.1) and using absolute coefficients gives

$$|\Delta N_q| < 0.000405, \quad |\Delta N_{F_2}| < 37.959, \quad |\Delta N_{\mathcal{C}_4}| < 31,549,532. \quad (\text{B.5})$$

For $x \geq 5$, (B.3) gives

$$|a_j| \leq 2^{j+2}x^j, \quad |e_j| \leq 2^{j+2}3^{2j+4}x^je^{-8x}.$$

The raw weights of N_q, N_{F_2}, N_{C_4} are respectively 2, 6, 12. Thus every perturbation majorant is a positive multiple of x^we^{-8x} , or a faster-decaying term, with $w \leq 12$; it decreases for $x \geq 5$. A single rational Taylor bound at $x = 5$ gives

$$|\Delta N_q| < 6.49 \cdot 10^{-11}, \quad |\Delta N_{F_2}| < 0.03, \quad |\Delta N_{C_4}| < 418,287. \quad (\text{B.6})$$

The bounds (B.5) and (B.6) are strictly smaller than (B.2). Since $b_0 > 0$, (B.1) proves all three signs for $u \geq 0$. Evenness proves Lemma 4.1.

C Endpoint audit of the transport cancellation

This appendix records the algebra suppressed by the phrase “simplifying the endpoint expression.” Put

$$z = p + L, \quad w = z + R, \quad a = \frac{Q(z) - Q(p)}{L}, \quad b = \frac{Q(w) - Q(z)}{R}.$$

Direct integration of $2 - Q''$ gives

$$\delta = \frac{L + Q'(p) - a}{L}, \quad \Lambda = L + R + a - b. \quad (\text{C.1})$$

Define

$$J_y = y^3 - 3yQ(y) + Q(y)Q'(y), \\ H_p = 3p^2 - 3Q(p) - 3pQ'(p) + Q'(p)^2 + Q(p)Q''(p),$$

and, for endpoint data,

$$U(c, d) = Q(c) + Q(d) - \frac{Q(d)Q'(d) - Q(c)Q'(c)}{d - c} + \frac{(Q(d) - Q(c))^2}{(d - c)^2}.$$

With $I_L = J_z - J_p$ and $I_R = J_w - J_z$, the identity needed in Lemma 9.1 is exactly

$$\begin{aligned} & \frac{-(J_z - J_p)/L + (J_w - J_z)/R}{\Lambda} + \frac{LH_p - (J_z - J_p)}{L^2\delta} \\ &= \Lambda + Q'(p) - a + \frac{U(p, z) - U(z, w)}{\Lambda} + \frac{U(p, z) - Q(p)(2 - Q''(p))}{L\delta}. \end{aligned}$$

To check it, substitute (C.1), replace z by $p + L$ and w by $p + L + R$, and multiply by the common denominator $L^2R\delta\Lambda$. The terms containing $Q'(w)$, then $Q'(z)$, then $Q''(p)$ cancel pairwise; the remaining endpoint values and position terms cancel after inserting $a = (Q(z) - Q(p))/L$ and $b = (Q(w) - Q(z))/R$. No differential equation for Q is used, so the identity holds for arbitrary smooth Q .

D Proof provenance

The maintained source map is `paper/PAPER_MAP.md`. The principal repository proof records are:

Claim	Proof record
Global $q, F_2, \mathcal{C}_4$, order three	<code>sources/riemann-signs-analytic-certificate.md</code>
Quotient and three-point equivalence	<code>sources/pf4-wronskian-reduction-certificate.md</code>
Transport identity and completion	<code>sources/pf4-transport-kernel-certificate.md</code>
Origin order-five obstruction	<code>sources/pf5-origin-certificate.md</code>
Continuous strict-PF ₄ Fourier separator	<code>sources/continuous-pf4-separator-certificate.md</code>

The corresponding MIND identities are

$$R140, R141, R147, R149, \quad R154\text{--}R156, \quad R153, R164, R171\text{--}R173, \\ R14, R37, R50, R72, \quad R165\text{--}R170.$$

The active replay boundaries are

$$CERT12, CERT5, CERT9, CERT11, CERT10.$$

The former CERT2 and CERT3 Arb boundaries remain archived historical evidence. These identifiers are evidence anchors, not bibliographic citations.

E Exact separator audit

This appendix records the algebraic boundary behind (12.7). Starting from the cumulants $\ell_2, \dots, \ell_6$, the verifier reconstructs the central moments

$$m_0 = 1, \quad m_1 = 0, \quad m_2 = \ell_2, \quad m_3 = \ell_3, \\ m_4 = \ell_4 + 3\ell_2^2, \quad m_5 = \ell_5 + 10\ell_3\ell_2, \\ m_6 = \ell_6 + 15\ell_4\ell_2 + 10\ell_3^2 + 15\ell_2^3,$$

and then forms $\det[m_{i+j}]_{i,j=0}^3$ directly. This independently regenerates the thirteen-term $\mathcal{C}_4$ polynomial; it does not import the expansion used for the Riemann kernel.

The recurrence (12.5) is then applied over $\mathbb{Q}[t]$. Substitution of (12.6) shows that each of the thirteen determinant monomials has common denominator P^{12} . After collecting the numerator H , exact rational arithmetic verifies

$$\deg H = 72, \quad [t^j]H = [t^{72-j}]H > 0 \quad (0 \leq j \leq 72),$$

with

$$\min_{0 \leq j \leq 72} [t^j]H = \frac{3}{65536}.$$

For a printable certificate, write

$$2^{54}H(t) = \sum_{j=0}^{72} a_j t^j, \quad a_j = a_{72-j}.$$

The 37 independent integer coefficients are:

j	a_j	j	a_j
0	824633720832	19	6987741510406573876435979520
1	48712961228800	20	17092065810580803244866465328
2	1429432562155520	21	39268975295924090460974938880
3	27770895893217280	22	84898287500864044148851126266
4	401708065273655296	23	173000552553192254015824625680
5	4613079582269972480	24	332750476261206454685845998080
6	43792294989734840832	25	604872001890033837007447191280
7	353349130016052428800	26	1040322081625845770286033564593
8	2472885380960577236992	27	1694567248719256234936725175440
9	15243207814279730964480	28	2616435044092694236816009466532
10	83763600430517403650816	29	3832176384214979954574258944560
11	414327966630038761815040	30	5327751824843881176247055457873
12	1859451652398640291313152	31	7034632066783174173709968967280
13	7621494792057501544629760	32	8825381090424751586774701723088
14	28689460125397617502291392	33	10523927205327676216804189288720
15	99652010444258917615558400	34	11931512643001413047042570621636
16	320698275507551388688130816	35	12863883307375841645695614141760
17	959582391519228973730931840	36	13190420295966739733605724579768
18	2677774483831421514536291072		

This table is generated by `scripts/generate_separator_coefficients.py`. Its machine-readable companion `generated/separator-coefficients.json` has SHA-256 prefix `87af00802c0929ca`; the complete hash is recorded in the release manifest. Coefficient positivity is a proof on the full domain $t > 0$, not a sampled test. Exact spot reconstructions at $t = 1/3, 1, 4$ independently compare the central determinant with H/P^{12} .

For the Fourier step, the identities

$$w + w^{-1} = u, \quad w^2 + w^{-2} = u^2 - 2, \quad w^3 + w^{-3} = u^3 - 3u$$

give (12.9) directly. Put

$$r = e^{1/64}, \quad x = r^2 = e^{1/32}.$$

Exact expansion gives

$$\text{disc}(R_c) = 4r^{10}G(x),$$

where

$$\begin{aligned} G(x) = & 432x^{13} - 4968x^9 + 900x^8 + 16200x^6 + 19044x^5 \\ & - 72600x^4 + 20000x^3 + 62100x^2 - 54334x + 13225. \end{aligned}$$

Writing $y = x - 1$ gives

$$\begin{aligned} G(1+y) = & -1 - 10y - 12y^2 + 680y^3 + 11532y^4 + 96660y^5 \\ & + 365400y^6 + 569664y^7 + 512172y^8 + 303912y^9 \\ & + 123552y^{10} + 33696y^{11} + 5616y^{12} + 432y^{13}. \end{aligned}$$

Moreover $0 < y < 1/30$. Indeed,

$$e < 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{18} = \frac{49}{18} < \frac{11}{4},$$

where $1/18 = (1/24) \sum_{j \geq 0} 4^{-j}$ bounds the tail beginning with $1/4!$. The binomial theorem gives

$$\frac{11}{4} < 1 + \frac{32}{30} + \frac{\binom{32}{2}}{30^2} + \frac{\binom{32}{3}}{30^3} = \frac{1891}{675} < \left(\frac{31}{30}\right)^{32}.$$

The sum of the positive terms at $y = 1/30$ is exactly

$$\sum_{j=3}^{13} \frac{[y^j]G(1+y)}{30^j} = \frac{1621193195302591}{36905625000000000} < 1.$$

The two omitted terms $-10y - 12y^2$ are negative. Consequently

$$G(1+y) < -1 + \sum_{j=3}^{13} \frac{[y^j]G(1+y)}{30^j} < 0,$$

and hence $\text{disc}(R_c) < 0$ without numerical evaluation.